Handout for Laboratory 1 : Setting up the home directory for the First Laboratory

Step 1: Login into the Linux server, babbage / hpc, using your roll number and assigned password :

```
$ ssh roll@server-ip
  <enter-your-initial-password>
```

Step 2: Set a new password and save it for use during this entire semester.

```
$ passwd
  <enter-new-password>
```

Caution : DO NOT share your password with any one. Violation can be easily detected and may result in disciplinary action.

Step 3: Set up your home directory for this course and create

3.1 Check your home folder using command "pwd"

```
$ pwd
/home/roll-number
```

3.2 Create a new folder (called a directory in Unix / Linux) for this course using command "mkdir", let the name of the folder be "CDLAB-MO2026".

```
$ mkdir CDLAB-MO2026
```

3.3 Move to the designated directory using command "cd".

```
$ cd CDLAB-MO2026
```

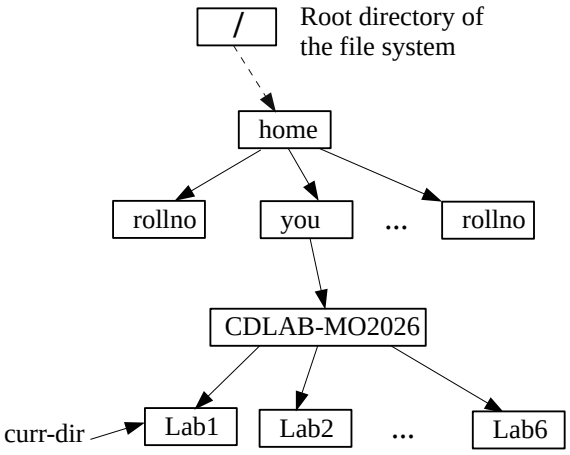
3.4 Create directories for Lab sessions using "mkdir". Let us create 6 directories for the first six laboratory sessions. We can create more than one directories by separating their names with one or more blank spaces.

```
$ mkdir Lab1  Lab2  Lab3  Lab4  Lab5  lab6
```

3.5 Move into the Lab1 directory to start working on the problems of the first lab session using the command "cd". Check that current directory has indeed changed by using "pwd".

```
$ cd Lab1
$ pwd
```

Your directory structure at this point in time, as built using the commands above is drawn below. A child directory of a directory exists at a deeper nesting level than that of its parent directory. A directory is represented by a rectangle and a regular file by a circle.



The current directory, denoted as "curr-dir", is set to "Lab1" after Step 3.5 has been completed.

Step 4: Attributes of Resources in the Kernel Data Structure

A file or a directory resource in the Linux/ Unix system has several attributes associated with, which are present in the kernel data structure of a directory. The "ls" command displays several attributes from the kernel directory data structure, but not all, in a human readable form. The "ls" command has several useful switches and we explain here the form "ls –al", where " –l" stands for long form description and "–a" stands for all entries of the argument directory.

4.1 If no directory argument is given to "ls –al", then the current directory contents are displayed, otherwise the directory given as argument to "ls" is displayed. Let the current directory be, /home/you/CDLAB-MO2026. Consider the sequence of commands and their outputs as shown below.

```
$ pwd
 /home/you/CDLAB-MO2026
$ ls – al
total 32
drwxrwxr-x 8 you you 4096 Jul 29 08:42 •
drwxrwxr-x 3 you you 4096 Jul 29 08:42 ••
drwxrwxr-x 2  you you 4096 Jul 29 08:42 Lab1
drwxrwxr-x 2  you you 4096 Jul 29 08:42 Lab2
drwxrwxr-x 2  you  you 4096 Jul 29 08:42 Lab3
drwxrwxr-x 2  you  you 4096 Jul 29 08:42 Lab4
drwxrwxr-x 2  you  you 4096 Jul 29 08:42 Lab5
drwxrwxr-x 2  you  you 4096 Jul 29 08:42 Lab6
```

4.2 Explanation for each line of the display generated by "ls –al".

total 32

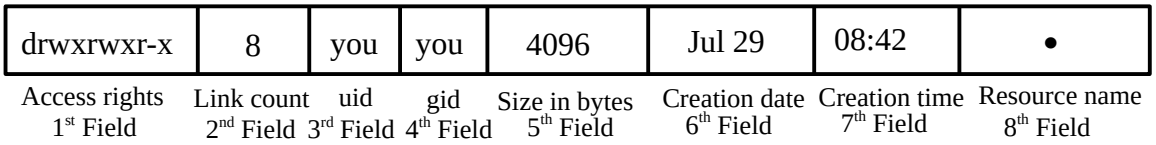
The line above indicates that the data structure for curr-dir, "/home/you/CDLAB-MO2026", occupies 32 bytes. This is because there are 8 entries in this directory, each line that follows the 1st one specifies one directory, and each entry requires 4 bytes.

4.3 The next 8 lines, list a directory per line, and the fields of a directory in a given line are separated by a blank space, as shown below.

drwxrwxr-x 8 you you 4096 Jul 29 08:42 •

These 8 fields are explicated in the following figure.

Figure L1.1 : File / Directory Attributes displayed by ls



Each field is explained with illustration in the following.

4.4 First field : Type of resource : directory and regular file

Composition	Type of resource 1st character	Access rights for owner (**u**) : 2nd to 4th character	Access rights for group owner (**g**) : 5th to 7th character	Access rights for others (**o**) : 8th to 10th character
Symbols used	d ≡ directory	rwx	rwx	r–x
	– ≡ regular file	r ≡ read access to the resource w ≡ write access to the resource x ≡ execute access for file and search access for a directory		
Interpretation	Current resource denoted by "•", in the 8th field, is a directory	Owner has all 3 rights : read from, write into and search the dir "•"	Group Owner has all 3 rights : read from, write into and search the dir "•"	All other users on the syatem has rights to read and search the dir "•" but can not write into dir "•"

Figure L1.2 : File / Directory Access Permissions

4.5 2nd Field : Link count : Number of times a resource is listed in the file system.
A directory resource is the 1st entry listed in its contents, the reason for the +1 in the calculation. It is also listed in the contents of its parent directory, and all its child directories list this directory as its parent in the 2nd entry of their contents. Note that for a file system that is a tree, there is exactly one parent for every directory node, however for a DAG(Directed Acyclic Graph) file system, the number of parents may be ≥ 1.

Link count	**Resource name**	**Parent Directory**	**Child Directories**
Number of parent directories + Number of child directories + 1 =1 + 6 + 1 = 8	"•" alias for the directory /home/you/CDLAB-MO2026	1 /home/you/	6 Lab1 Lab2 Lab3 Lab4 Lab5 Lab6
1 + 1 + 1 = 3	•• alias for the directory /home/you	1 /home/you	1 /home/you/CDLAB-MO2026
1+1 = 2	Lab4 alias for the directory /home/you/CDLAB-MO2026/Lab4	1 /home/you/CDLAB-MO2026	0

Figure L1.3 : Link Count of a File / Directory

The calculation of link count field for the following 3 directories are explained above.

drwxrwxr-x 8 you you 4096 Jul 29 08:42 •
drwxrwxr-x 3 you you 4096 Jul 29 08:42 ••
drwxrwxr-x 2 you you 4096 Jul 29 08:42 Lab4

Exercise 2. Calculate manually the link count of various resources, files / directories, in your home directory and compare the count with that displayed by "ls – l".

4.6 Third and Fourth Fields : Displays the identity of owner and group of owner.
Every file resource in Linux is assigned two owners by default – one is the identity of the user who is the creator of the resource, denoted as "uid", and the other is the identity of the group, denoted by

"gid", to which the user "uid" belongs to. For example, "uid" for a student is denoted in this handout as "you", while "gid" of a student is also denoted as "you", which may be different in your system. In practice, a student of your batch may have uid as her/his "roll-number" and gid as "btech3" indicating a 3rd year B Tech student. For a faculty member, the uid and gid respectively may be, "sb" and "fac", to represent "Supratim Biswas" and his group as "faculty". The gid specifies the privileges accorded to the different groups in the file system and every user in the group is constrained by these predefined privileges.

4.7 Fifth Field : Size of the directory in bytes.

A directory may be visualized as a tabular data structure, and the size of a directory determines the maximum number of entries in it. Assuming each entry of the directory occupies 4 bytes, 4096 denotes that the directory can have as many as 1024 entries (files or directories). Since the first 2 entries of every directory are reserved for "•" and "••", current and its parent directory, it implies that 1022 other distinct entries may exist in it. It may be noted that the actual internal data structure of a node in the Linux file system (directory or a file) has several other attributes, but "ls" is permitted to read only a few fields from this kernel data structure and display them.

The size 4096 is used here as the default size allocated to every directory on creation, that is empty except the first 2 mandatory entries. However this is not a limitation, since when an user has adds more than 1024 entries in a certain directory, the OS allocates another chunk of 4096 bytes, to make it 8192 bytes, in order to let it grow further (though having a large monolithic organization is not considered a good design).

4.8 Sixth and Seventh Fields : Date and time of creation of a resource

The time stamp of a resource is a field in the internal kernel data structure. Not only the time stamp of a resource at the point of creation is saved, time stamps related to latest access and last modifications of the resource are also preserved. The "ls –l" command displays the creation time stamp only, which comprise the date, month and year of creation, along with the time in hours and minutes, for every resource in the directory.

For resources recently created, say within the last 6 months, it does not show the year while for older ones it shows the year but suppresses the time. Consider the following displays from "ls –l" from some directory.

```
drwxrwxr-x  2 you you 4096  Jul 29   08:42  Lab4
drwxr-xr-x   2 sb   sb    4096  Feb 26  2021  Templates
–rw-rw-r--   1 sb   sb         71 Sep 20  2024  save.txt
```

The information conveyed by the above lines in the given order are summarized below.

- Name of the resource is "Lab4"; it is a directory; owner (u) and group (g) have all the three permissions, other-users can only read and search Lab4 but cannot write into it; link count is 2; uid and gid are both "you"; size is 4096; was created on July 29 at 08:42 hours (in 2026 not shown).
- Name of the resource is "Templates"; it is a directory; owner (u) has all read and write access it is not an executable file, group members (g) have read and write access, while other members

have only access; link count is 1; uid and gid are both "sb"; size is 4096; was created on February 26 in 2021 (time not shown).

- Name of the resource is "txt.save"; it is a regular file; owner (u) has all 3 accesses, group members (g) and other members (o) can read and search but cannot write; link count is 2; uid and gid are both "sb"; size is 71 bytes; was created on September 20 in 2024 (time not shown).

4.9 Eighth Field : Name of the resource, file or directory.

Exercise 3. For a C program, saved in a file named as "prog1.c" in your home-dir, with 225 characters, created on July 30, 2026 at 14:20 hrs, write the display as would be displayed by "ls – l prog1.c".

4.5 Secure your home directory

The home directory of an user (referred to as home-dir, "/home/roll-no") can be secured to ensure that this directory can not be accessed by any other user in the system, except the administrator. This will prevent copying of resources in the file system rooted at the home-dir of an user, intentionally or otherwise, by other users in the system. This is possible if we know how to change the default access permissions set by your OS. In Linux OS, the command "chmod", does this task. The usage of chmod is

chmod <person> <new-perms> <name>

The candidates for <person> are u(user), g(group), o(others). Any subset of {u, g, o} may be given. The candidates for <new-perms> are defined by + (to add) or – (to remove) a permission, followed by one or more of read(r), write(w), execute(x). The <name> attribute is the the name of the resource whose permissions need to be changed.

Example : Let the permission of "/home/you" be as given below.

drwxrwxr-x 3 you you 4096 Jul 29 08:42 /home/you

i) The command

$ chmod g–w /home/you

specifies that the write access for group(g) is to be removed for the directory /home/you. This command changes the permissions as shown below.

drwxr-xr-x 3 you you 4096 Jul 29 08:42 /home/you

(ii) To add write permission to both g and o, we execute the command :

$ chmod go+w /home/you

which changes the listing to

drwxrwxrwx 3 you you 4096 Jul 29 08:42 /home/you

(iii) To prevent others, both g and o, to access /home/you, we issue the command :

$ chmod go–rwx /home/you

which results in

drwx------ 3 you you 4096 Jul 29 08:42 /home/you

End of Handout